SEEDLINE · USER REQUIREMENTS

SeedLine user requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SDL-URS-001	Version	0.1
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
Source	docs/03a-user-requirements.md @ 0a57933		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 pipeline, kit 1.9.1 upgrade): user requirements drawn from the users, stakeholders and constraints in SDL-PRB-001

SeedLine user requirements

These are the needs of the people who would use, build and support SeedLine, written from their side. They come from the users and stakeholders in SDL-PRB-001 (Tables 1 and 2) and its constraints; none has yet been checked with a co-design partner, so every priority is a proposal for review (SDL-PRB-001, "User research and co-design"). The product requirements that meet them are in SDL-PRS-001 (`03-requirements.md`), and the links are generated in SDL-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	One seed per station	Each planting station gets one seed, so seed is not wasted and the stand does not need thinning	High	Smallholder farmer; market gardener (PRB Table 1)
UR-002	Performance	Even spacing	Plants come up evenly along the row, without the clumps and gaps of hand dropping	High	Smallholder farmer; extension worker (PRB Table 1; Nielsen, Purdue)
UR-003	Performance	Spacings for local crops	The seeder sows the in-row spacings of the crops grown, maize, beans, sorghum and groundnut first and vegetables later, from about 25 to 400 mm	High	Smallholder farmer; market gardener (PRB Table 1; SDL-DDR-001 item 2)
UR-004	Performance	Spacing independent of pace	Spacing stays the same when the operator walks faster or slower or stops	Medium	Smallholder farmer (PRB)
UR-005	Usability	Crop change by a printed part	A crop or seed size is changed in the field by swapping a part a cooperative or local maker can print on a common desktop printer	High	Cooperative or agro-dealer with a printer (PRB Table 1)
UR-006	Performance	Right sowing depth	Seed is placed at the depth the crop needs and held there along the row	High	Smallholder farmer; extension worker (PRB)
UR-007	Performance	Faster than the hoe	Planting a plot takes much less labour than hoe or jab planting, so it fits in the short planting window	High	Smallholder farmer (PRB, about 56 h/ha by hoe)
UR-008	Usability	One person can push it	One person can push the seeder at walking pace on a tilled seedbed	High	Smallholder farmer, often one worker (PRB constraints)
UR-009	Usability	Light and fits the operator	One person can lift and turn it at a row end, and the handle suits the operator's height	High	Smallholder farmer (PRB)
UR-010	Usability	Seed capacity and easy filling	It holds enough seed for long runs, fills from a jug without spilling and empties quickly for a crop change	Medium	Smallholder farmer; market gardener (PRB)
UR-011	Usability	Next row marked	The next row is marked at the chosen row spacing, so rows come out straight and parallel	Medium	Smallholder farmer (SDL-DDR-001 item 3)
UR-012	Safety	No pinch points or sharp edges in use	Hands, feet and clothing cannot be caught by moving parts, and there are no exposed sharp edges	High	Every user (PRB; PRC Safety)
UR-013	Environment and durability	Lasts several seasons	Frame, drive and opener last several seasons on small plots with only wear parts replaced	Medium	Smallholder farmer; cooperative (PRB)

ID	Category	Title	Need	Priority	Source
UR-014	Cost	Prototype within the value-engineering target	The proof-of-concept prototype's parts stay within the hypothetical USD 400 value-engineering target	Medium	Amish Chadha, `project.yaml` budget_usd (STANDARDS section 18)
UR-015	Cost	Affordable to replicate	A replicated base seeder costs no more than a low-cost plate seeder, about USD 200 in parts	High	Smallholder farmer; cooperative (PRB constraints)
UR-016	Deployment and assembly	Built and repaired locally	A local workshop can build and repair it from common steel sections, bicycle or roller chain parts and bearings, with a drill and spanners or a welder	High	Local welder or maker (PRB Table 1)
UR-017	Safety	Treated seed handled safely	Seed dressed with pesticide can be filled, emptied and cleaned out with little contact	Medium	Smallholder farmer; extension worker (PRB operating environment; PRC Safety)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.